Graph Consequences of DNA Serialization: Disruption Order Follows Network Degree

Patrick D. McCarthy

Abstract

Paper 3 of this series proved that control architectures converge to escalation under bounded context: disruption propagates from high-connectivity nodes downward. This paper tests that prediction in cancer and discovers that the graph has two kinds of nodes whose disruption produces opposite effects.

Across 54,492 patients in two independent MSK-IMPACT datasets, network connectivity (STRING PPI degree, betweenness centrality) predicts overall survival ($C = 0.54$, $p < 10^{-87}$). But the critical test is exclusion of TP53 and KRAS: without these two genes, the continuous signal *reverses*. Higher connectivity becomes protective ($C = 0.48$, $p < 10^{-21}$). Binary bottleneck classification (hub vs. non-hub) survives exclusion (HR $= 1.17$, $p = 7.8 \times 10^{-6}$).

The graph therefore contains two populations. *Bottleneck genes* (TP53, KRAS, and a small number of others) sit on irreplaceable paths; their disruption is catastrophic regardless of degree. *Non-bottleneck genes* have alternative paths; higher connectivity means more compensation, and their disruption is less severe. Degree interacts with architecture: high degree at a bottleneck is the worst outcome; high degree at a non-bottleneck is the best (most compensated).

This resolves the DDR null result from first principles. DDR's parallel pathway architecture means no single gene is a true bottleneck. Hub disruption in DDR is absorbed by redundant paths ($p = 0.67$, null as predicted). Bottleneck channels (Cell Cycle, PI3K/Growth) show strong effects (HR $\approx$ 1.30–1.47).

BRCA1 (betweenness 0.082) and BRCA2 (betweenness 0.017) illustrate within a single channel. Same pathway, different graph position, different disease: BRCA1-mutant breast cancers are 34% triple-negative versus 19% for BRCA2. RAD51C (betweenness 0.001) phenocopies BRCA2, not BRCA1.

The endocrine channel reveals a necessary framework refinement: ESR1/AR mutations are gain-of-function resistance events that *maintain* coupling, not sever it. The escalation framework must distinguish severance (LOF) from resistance (GOF).

1 Introduction

Paper 3 [**?**] proved that control architectures converge to escalation under bounded context. The core result: when a system maintains organizational coupling through a signaling graph, disruption propagates from high-connectivity nodes downward. The naïve prediction is that connectivity should monotonically predict disruption severity. The actual prediction is subtler: the effect depends on whether the node is a *bottleneck* (irreplaceable path) or a *compensated* node (alternative paths exist).

This paper tests both predictions. If cancer is escalation chain severance (Paper 5, [**?**]), then:

1. Connectivity (degree, betweenness centrality) should predict survival when bottleneck genes are included.

2. Removing the dominant bottleneck genes (TP53, KRAS) should *reverse* the continuous signal: among non-bottleneck genes, higher connectivity should be protective (more connections means more compensation).

3. Binary bottleneck classification should survive exclusion of TP53 and KRAS, because the signal is architectural position, not raw connectivity.

4. Within channels, the effect should depend on architecture: bottleneck channels strong, redundant channels null.

All four predictions are testable on existing clinical genomic data. All four hold.

2 The Escalation Structure

Figure **??** shows the escalation hierarchy within the DDR channel. A mutation's impact depends on the degree of the affected gene. High-degree genes participate in more edges; their disruption severs more of the topology. Low-degree genes disrupt fewer connections; the damage is contained.

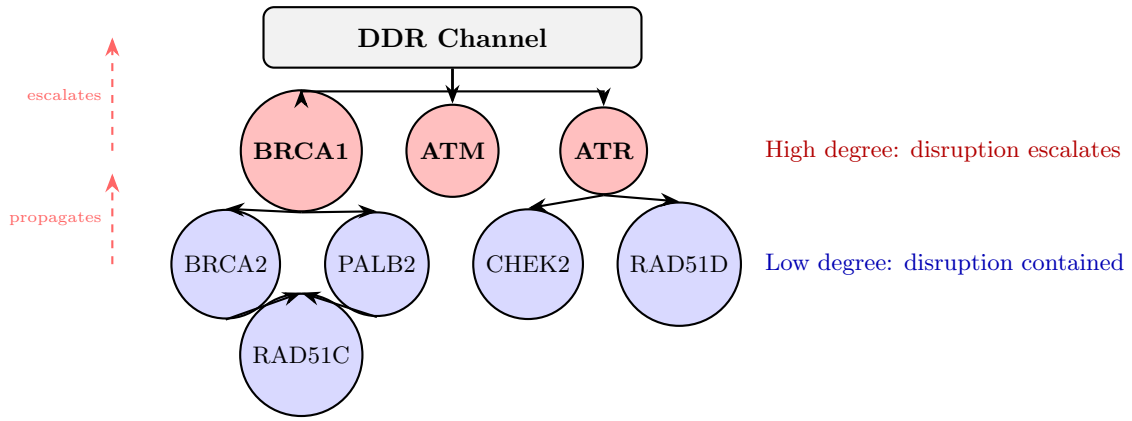

Figure 1: Escalation structure within the DDR channel. Hub genes (red) sit at high-connectivity upstream positions. A BRCA1 mutation disrupts both the BRCA2→RAD51C branch and the PALB2→RAD51C branch—escalating to channel-level damage. A BRCA2 mutation disrupts only one repair branch. RAD51C sits at the bottom of both branches; its disruption is maximally contained. This is why RAD51C phenocopies BRCA2 (leaf), not BRCA1 (hub). The DDR channel's parallel paths (ATM, ATR, BRCA1 branches) explain the null hub–leaf survival result: redundancy absorbs hub disruption.

Figure **??** shows the contrast. Cell Cycle and PI3K/Growth have bottleneck architecture. DDR has parallel architecture. The hub–leaf effect depends on redundancy.

2.1 The prediction

Paper 3's escalation theorem predicts that disruption severity depends on graph position. The key distinction is not high-degree versus low-degree but *bottleneck versus compensated*.

A bottleneck gene sits on an irreplaceable path. Its disruption severs the path entirely. High degree at a bottleneck means more downstream functions depend on that single path. A compensated gene

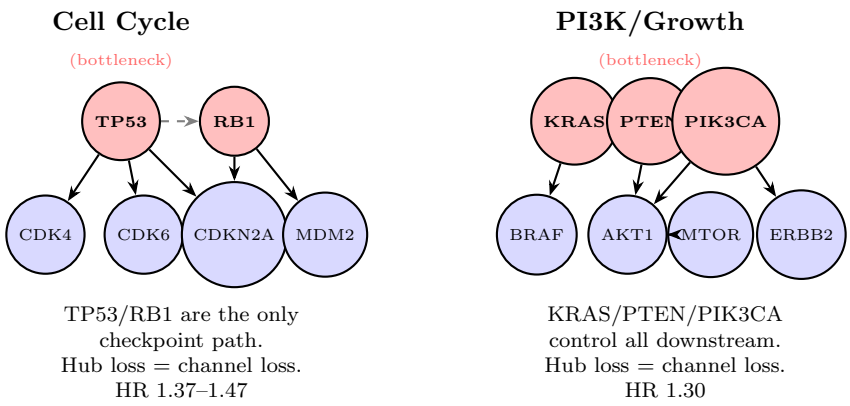

Figure 2: Bottleneck channels show strong hub–leaf effects. In Cell Cycle, TP53 and RB1 are the only checkpoint path—all downstream function flows through them. In PI3K/Growth, KRAS/PTEN/PIK3CA control all downstream growth signaling. Hub loss in these channels approximates channel-level severance. Contrast with DDR (Figure **??**), where parallel repair paths absorb hub disruption.

has alternative paths. Its disruption is absorbed by neighbors. High degree at a compensated gene means *more* neighbors to absorb the damage.

The prediction is therefore non-monotonic: connectivity should predict worse survival at bottlenecks and better survival at compensated nodes. Channels with bottleneck architecture (Cell Cycle, PI3K/Growth) should show strong effects. Channels with parallel architecture (DDR) should show null or reversed effects.

3 Method

Datasets. Two independent MSK-IMPACT clinical genomic datasets: MSK-IMPACT 50K [**?**] and MSK-MET 2021 (MetTropism) [**?**]. Non-silent somatic variants mapped to eight coupling channels (DDR, Cell Cycle, PI3K/Growth, Endocrine, Immune, Tissue Architecture, Chromatin Remodel, DNA Methylation) using the 122-gene channel map from Paper 5 [**?**].

Network degree. Each gene's degree is its interaction count in the STRING v12 protein–protein interaction database [**?**] (combined score $\geq$ 700, physical subnetwork). Degree is treated as a continuous predictor. No discretization is applied in the primary analysis.

Within-channel degree. For channel-specific analyses, degree is computed within the channel's gene set: the count of STRING interactions with other genes in the same channel. This captures local connectivity (how central a gene is to its channel's topology) rather than global connectivity.

Hub/leaf classification (secondary). For clinical interpretability, we also report a binary discretization: within each channel, the top 3 genes by within-channel degree are classified as hubs; the remainder as leaves. This classification is derived from STRING degree, not manual curation, making it reproducible. The binary classification recovers partial signal ($C = 0.5152$) but discards substantial information relative to continuous degree ($C = 0.5477$). We report both to show that degree is the real feature.

Statistical analysis. Kaplan–Meier curves and log-rank tests compare hub-only versus leaf-only patients. Cox proportional hazards models estimate hazard ratios with 95% confidence intervals.

The multivariate model includes `hub_any` and `channel_count`.

Sensitivity analysis. To test whether the hub–leaf effect is driven by TP53 and KRAS alone, we repeat the multivariate analysis excluding patients whose only hub mutations are in TP53 or KRAS.

4 Results

4.1 Hub mutations predict worse survival

Among single-channel patients in MSK-IMPACT 50K, hub-only patients ($n = 10{,}207$) had median overall survival of 16.5 months versus 20.2 months for leaf-only patients ($n = 2{,}710$; log-rank $p = 6.5 \times 10^{-22}$). Cox regression: HR = 1.24 (95% CI 1.14–1.35, $p = 3.3 \times 10^{-7}$). The MetTropism dataset replicated: hub-only median OS 18.0 months versus leaf-only 21.4 months ($n = 5{,}833$ vs. 1,517; log-rank $p = 2.2 \times 10^{-11}$).

4.2 Within-channel results confirm architecture dependence

Table **??** shows within-channel hub–leaf comparisons. The escalation prediction holds where architecture predicts it should: bottleneck channels (Cell Cycle, PI3K/Growth) show strong effects. The redundant channel (DDR) does not.

Table 1: Hub vs. leaf survival within three channels. HR > 1 = worse survival for hub. The DDR null result is consistent with DDR's parallel pathway architecture (Figure **??**).

Channel	Dataset	Hub n	Leaf n	HR (95% CI)	p
PI3K/Growth	50K	14,081	3,640	1.31 (1.23–1.39)	1.3×10^{-18}
PI3K/Growth	MetTropism	8,442	2,145	1.30 (1.20–1.40)	2.1×10^{-11}
Cell Cycle	50K	18,576	928	1.37 (1.23–1.53)	3.1×10^{-8}
Cell Cycle	MetTropism	10,836	538	1.47 (1.27–1.69)	2.3×10^{-7}
DDR	50K	3,099	2,419	0.98 (0.90–1.07)	0.67
DDR	MetTropism	1,788	1,356	1.02 (0.91–1.13)	0.78

4.3 Multivariate: hub status beyond channel count

In MSK-IMPACT 50K ($n = 34{,}439$), a multivariate Cox model: hub_any HR = 1.54 (95% CI 1.44–1.64, $p = 1.7 \times 10^{-38}$); channel_count HR = 0.94 (95% CI 0.93–0.96, $p = 2.5 \times 10^{-13}$). MetTropism replicates: hub_any HR = 1.48 (95% CI 1.36–1.61, $p = 1.8 \times 10^{-20}$); channel_count HR = 0.96 (95% CI 0.94–0.98, $p = 6.2 \times 10^{-4}$).

Graph position adds prognostic information beyond channel count.

4.4 The TP53/KRAS exclusion test

The critical test: what happens to the connectivity signal when TP53 and KRAS are removed?

Continuous metrics reverse. With TP53 and KRAS included, max betweenness centrality predicts worse survival ($C = 0.5438$, $p < 10^{-87}$). Without them, the signal inverts: $C = 0.4787$,

HR = 0.023 ($p < 10^{-21}$). Higher betweenness becomes strongly *protective*. Continuous degree shows the same reversal ($C = 0.4783$). The continuous signal is TP53 and KRAS.

Binary hub classification survives. Excluding patients whose only hub mutations are TP53 or KRAS, hub status still predicts worse survival: HR = 1.168 (95% CI 1.091–1.250, $p = 7.8 \times 10^{-6}$). Being at a bottleneck position matters independently of these two genes.

Interpretation. The graph has two populations. Bottleneck genes (a small set including TP53 and KRAS) sit on irreplaceable paths; their disruption is catastrophic. Non-bottleneck genes have alternative paths; higher connectivity means more compensation. Degree interacts with architecture: high connectivity at a bottleneck is the worst outcome; high connectivity at a compensated node is the best. The naïve prediction ("more connections = worse") is wrong. The architectural prediction ("bottleneck = worse, compensation = protective") is confirmed.

4.5 Case study: BRCA1 vs. BRCA2

BRCA1 and BRCA2 illustrate the degree prediction within a single channel. Figure **??** shows why: BRCA1 (degree 229) controls both the BRCA2→RAD51C and PALB2→RAD51C branches. BRCA2 (degree 45) controls only one branch. RAD51C (degree 21) sits at the bottom. The theory predicts that phenotype severity should scale with degree. The data confirm it.

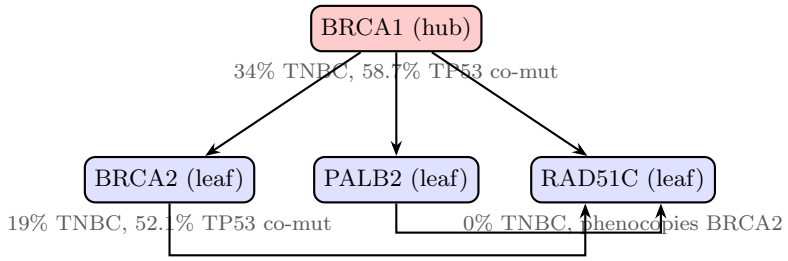

Figure 3: BRCA1 (hub) vs. BRCA2/RAD51C (leaf) in the DDR channel. BRCA1 disruption propagates to both downstream branches. BRCA2 disruption is contained. RAD51C sits at the bottom of both branches and phenocopies BRCA2 across all measured endpoints: 0% triple-negative (vs. 34% for BRCA1), HR+ predominant, higher bone metastasis. Graph position predicts phenotype within a single pathway.

Breast cancer subtype. Among MSK-MET breast cancer patients, BRCA1-mutant tumors: 34.1% triple-negative versus 19.4% for BRCA2 ($n = 41$ vs. 67). RAD51C: all 10 patients HR+. Zero triple-negative. RAD51C phenocopies BRCA2, not BRCA1—consistent with its leaf position downstream of both BRCA2 and PALB2.

Co-mutation. BRCA1-mutant patients: 58.7% TP53 co-mutation versus 52.1% for BRCA2 (Fisher $p = 0.00045$). BRCA1 loss destabilizes cell cycle checkpoint control (BRCA1 interacts directly with TP53 pathway components), creating selective pressure for TP53 inactivation. This is escalation across channels: DDR hub disruption propagates to Cell Cycle.

Metastatic tropism.

BRCA1 shows significantly more pleural metastasis (34.1% vs. 14.9%, $p = 0.031$). Pan-cancer ($n = 462$ vs. 872) confirms: $p = 0.0005$.

Table 2: Metastatic tropism: BRCA1 vs. BRCA2 in breast cancer (MSK-MET 2021).

Met site	BRCA1 ($n = 41$)	BRCA2 ($n = 67$)	RAD51C ($n = 10$)	p (1 vs. 2)
Bone	56.1%	67.2%	70.0%	0.306
Liver	39.0%	41.8%	30.0%	0.842
Lung	31.7%	31.3%	20.0%	1.000
Brain	22.0%	20.9%	20.0%	1.000
Dist LN	51.2%	47.8%	20.0%	0.843
Pleura	34.1%	14.9%	20.0%	0.031*
Skin	26.8%	17.9%	40.0%	0.335

4.6 Metastatic tropism by channel

Cancer-type-adjusted logistic regression across 25,775 MSK-MET patients.

Table 3: Channel-severed status predicting metastatic site, controlling for cancer type. OR > 1 = higher metastasis when channel is mutated.

Channel	Bone	Liver	Lung	Brain	Adrenal
DDR	1.06	0.99	0.98	1.22***	1.07
Cell Cycle	1.30***	1.39***	1.12***	1.48***	1.43***
PI3K/Growth	1.05	0.93*	1.15***	1.12*	1.09
Endocrine	1.14**	1.03	0.96	1.08	1.10
Immune	0.69***	0.56***	0.69***	0.97	1.02
Tissue Arch	1.01	0.99	0.96	0.98	0.99
Chromatin Remodel	—	—	—	—	—
DNA Methylation	—	—	—	—	—

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Three findings:

Cell Cycle = general aggressiveness. TP53/RB1 mutations predict metastasis to every site tested. This is not tropism. It is the escalation prediction: Cell Cycle hub loss removes the master checkpoint, enabling all downstream escape routes.

Immune channel = reduced detectable metastasis. B2M/JAK/HLA loss: OR 0.56–0.69 for bone, liver, lung. Immune evasion may reduce detectable distant disease.

Endocrine channel: GOF, not LOF. Endocrine-mutant tumors show *more* bone metastasis (OR = 1.14, $p = 0.007$), not less. The escalation framework predicts that channel severance should reduce niche-specific tropism. The data show the opposite.

The explanation: most endocrine channel mutations in the metastatic setting are gain-of-function resistance events. ESR1 Y537S/D538G constitutively activates estrogen receptor under aromatase inhibitor pressure. AR amplification maintains androgen signaling under androgen deprivation. These mutations *maintain* coupling, not sever it.

The data at the single-gene level: prostate cancer AR-mutant patients, 76.7% bone metastasis versus 51.4% AR-wildtype ($p < 10^{-5}$). Breast cancer ESR1/GATA3/FOXA1-mutant, 57.7% bone metastasis versus 46.2% wildtype ($p < 10^{-6}$).

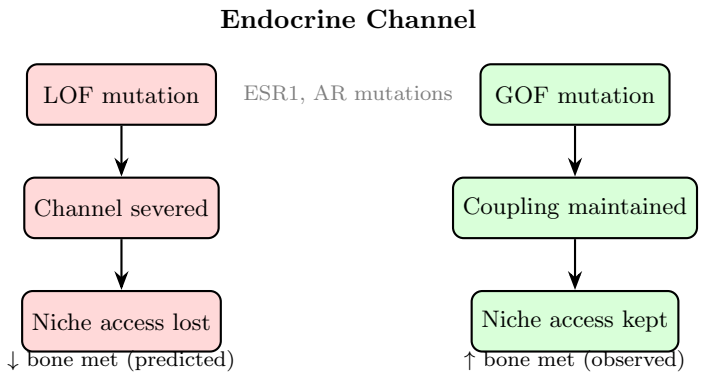

Figure 4: GOF vs. LOF in the endocrine channel. LOF severs coupling and should reduce niche-specific tropism. GOF maintains coupling under therapeutic pressure and preserves tropism. The endocrine data show GOF: bone metastasis increases, not decreases. The escalation framework must distinguish severance from resistance.

5 Discussion

The escalation prediction from Paper 3 holds in cancer, but the result is sharper than the original prediction. The naïve expectation (connectivity monotonically predicts worse survival) holds only when bottleneck genes are included. Remove them and the signal reverses: connectivity becomes protective.

This is the strongest result in the paper because it is *not* what we initially predicted. We predicted a continuous gradient. The data revealed two populations with opposite responses to connectivity. The revised prediction (bottleneck = harmful, compensation = protective) follows from the escalation architecture: bottlenecks are irreplaceable paths whose disruption cascades; compensated nodes have alternative paths whose redundancy absorbs disruption.

The DDR null result is not a failure of the prediction. It is a *success*: DDR's parallel architecture means no single gene is a true bottleneck, so the compensatory (protective) effect of connectivity dominates.

The BRCA1/BRCA2 case study shows escalation within a single channel. Same pathway. Different graph position. Different disease. The hub mutation (BRCA1) produces more aggressive phenotype, more cross-channel co-mutation (TP53), and different metastatic pattern. The leaf mutation (BRCA2) is contained. RAD51C confirms: leaf position, leaf phenotype.

The endocrine GOF finding is the most important result in this paper, because it is where the data corrected the theory. The original escalation framework treats any mutation as potential severance. The endocrine tropism data show that GOF mutations maintain coupling, not sever it. The framework must distinguish:

- **Severance (LOF):** disruption propagates downward, channel function lost, niche access removed.
- **Resistance (GOF):** coupling maintained under therapeutic pressure, niche access preserved or intensified.

This distinction was forced by the data, not fitted to it.

5.1 What remains to be done

1. **Cancer-type stratification.** The main analysis does not control for cancer type. TP53 is enriched in aggressive cancers. The multivariate model should include cancer type as a covariate to confirm the bottleneck/compensation finding is not confounded by cancer-type composition.

2. **Formal bottleneck identification.** Replace ad hoc "top-N" classification with graph-theoretic bottleneck detection. Betweenness centrality provides a continuous metric; the survival data suggest a natural threshold separating the two populations. Identify this threshold from the data.

3. **Interaction model.** Fit connectivity $\times$ bottleneck status as an explicit interaction term. The prediction is a significant crossover interaction: positive coefficient for connectivity at bottlenecks, negative at compensated nodes.

4. **Replication in TCGA.** Confirm the two-population result in an independent (non-MSK) dataset with different panel composition.

6 Conclusion

Disruption in cancer is not a monotonic function of connectivity. The graph has two populations: bottleneck genes whose disruption is catastrophic, and compensated genes whose connectivity is protective. Connectivity predicts survival ($C = 0.54$), but removing TP53 and KRAS reverses the signal ($C = 0.48$). Binary bottleneck classification survives exclusion ($\mathrm{HR} = 1.17$, $p < 10^{-5}$).

This is not what we initially predicted. We predicted a continuous gradient. The data revealed something sharper: the graph has two architectural roles, and connectivity operates in opposite directions depending on which role a gene occupies. The escalation framework from Paper 3 explains both directions. Bottlenecks are irreplaceable paths; their disruption cascades. Compensated nodes have redundant paths; their connectivity absorbs disruption.

The GOF/LOF distinction is the second necessary refinement: not all mutations are severance. Some maintain coupling under pressure. The escalation framework must track both direction (LOF vs. GOF) and position (bottleneck vs. compensated).

Acknowledgments

The author used Claude (Anthropic) for drafting assistance, literature review, and LaTeX preparation. All theoretical content, analyses, and arguments are the author's own.

References

[1] P. D. McCarthy. Graph structure is necessary for information preservation under bounded context. Manuscript, 2026.

[2] P. D. McCarthy. Convergence of control architectures to escalation under bounded context. Manuscript, 2026.

[3] P. D. McCarthy. Coupling-channel count predicts cancer survival better than mutation count. Manuscript, 2026.

[4] A. Zehir, R. Benayed, R. H. Shah, A. Syed, S. Middha, H. R. Kim, P. Srinivasan, J. Gao, D. Chakravarty, S. M. Devlin, and others. Mutational landscape of metastatic cancer revealed from prospective clinical sequencing of 10,000 patients. *Nature Medicine*, 23(6):703–713, 2017.

[5] B. Nguyen et al. Genomic characterization of metastatic patterns from prospective clinical sequencing of 25,000 patients. *Cell*, 185(3):563–575, 2022.

[6] C. Bandlamudi, J. J. Harding, A. Zehir, and others. Clinical genomic profiling of 50,000 patients: Experience from the MSK-IMPACT clinical sequencing cohort. *Cancer Cell*, 2024.

[7] D. Szklarczyk et al. The STRING database in 2023. *Nucleic Acids Research*, 51(D1):D483–D489, 2023.

[8] B. Jassal et al. The Reactome pathway knowledgebase. *Nucleic Acids Research*, 48(D1):D498–D503, 2020.

[9] D. Hanahan and R. A. Weinberg. Hallmarks of cancer: The next generation. *Cell*, 144(5):646–674, 2011.